Materials Discovery Model Report

Executive Summary

This report provides comprehensive information about the materials discovery workflow, including data sources, model architecture, performance metrics, and important limitations. Generated on 2026-03-25 17:05:07.

1. Data Source

Source: Materials Project (materialsproject.org)
Dataset Size: 1,000 materials
Material Types: Binary and ternary alloys, oxides, intermetallics
Data Types: Formation energies, band gaps, energy above hull, structural properties
Last Updated: Real-time data from Materials Project API

2. Model Architecture

Generative Model: Variational Autoencoder (VAE)
- **Input:** Material composition features (6 dimensions)
- **Latent Space:** 5-dimensional compressed representation
- **Decoder:** Reconstructs material properties from latent space
- **Training:** 50 epochs, Adam optimizer, KL divergence regularization

Predictive Model: Random Forest Classifier
- **Features:** 7 material properties (formation energy, band gap, E_hull, etc.)
- **Target:** Binary classification (synthesizable vs. not synthesizable)
- **Training Data:** 800 materials (80/20 train/test split)
- **Hyperparameters:** 200 trees, max depth 10, min samples split 5

Ensemble Method: Weighted combination of ML and rule-based predictions
- **ML Weight:** 70%
- **Rule-based Weight:** 30%
- **Final Prediction:** Ensemble probability > 0.5

3. Performance Metrics

Metric	Value

Accuracy	1.0000
Precision	1.0000
Recall	1.0000
F1-Score	1.0000
Cross-Validation Mean	1.0000
Cross-Validation Std	0.0000

4. Important Limitations

1. Binary Alloys Only: Current model trained primarily on binary alloy systems. Performance may be reduced for ternary and higher-order systems.

2. Thermodynamics vs. Kinetics: Model evaluates thermodynamic stability (energy above hull) but does not account for kinetic barriers to synthesis. A thermodynamically stable material may still be difficult or impossible to synthesize due to kinetic limitations.

3. Training Data Bias: Model predictions are only as good as the training data. Materials significantly different from the training distribution may have unreliable predictions.

4. Experimental Conditions: Model does not account for specific experimental conditions, equipment capabilities, or operator expertise. Actual synthesis success rates may vary.

5. Property Correlations: The model uses correlations between calculated material properties and experimental synthesizability. These correlations may not hold for all material systems.

6. Uncertainty Quantification: While the model provides confidence scores and calibration metrics, these should be interpreted as relative indicators rather than absolute probabilities.

5. Recommended Synthesis Workflow

Step 1: Review Predictions
- Examine ensemble probability (>0.7 recommended for first attempts)
- Check thermodynamic stability category
- Review calibration status and in-distribution classification

Step 2: Prioritize Candidates
- Use synthesis priority ranking for experimental order
- Consider cost-benefit analysis for resource allocation
- Start with high-confidence, high-priority candidates

Step 3: Method Selection
- Review recommended synthesis method and conditions
- Verify equipment availability and compatibility
- Consider alternative methods if recommended method unavailable

Step 4: Experimental Planning
- Calculate feedstock masses using provided wt% compositions

- Plan for multiple synthesis attempts (success rates vary)
- Document all experimental conditions and outcomes

Step 5: Validation and Iteration
- Compare experimental outcomes with model predictions
- Update local knowledge base with experimental results
- Use insights to improve future predictions

6. Generated Dataset Summary

Total Materials in Export: 2
Predicted Synthesizable: 2 (100.0%)
High Confidence Predictions (confidence > 0.8): 2 (100.0%)
In-Distribution Materials: 2 (100.0%)
Out-of-Distribution Materials: 0 (0.0%)